Configuration

Encoding Method

amplitude

Normalize input vector as quantum amplitudes (|psi> = x / ||x||).

Number of Qubits: 8

Batch Size: 64

Number of Batches: 100

Environment

CUDA: Available (A100)

QDP: Available

PennyLane: Available

Qiskit: Available

GPU connected. Using live benchmark data.

QDP: GPU-Accelerated Quantum Data Processing

Apache Mahout | Rust + CUDA | Zero-Copy DLPack Interop

Pipeline Demo | Framework Comparison | Scaling Analysis

End-to-End Encoding Pipeline

Pipeline Architecture

Workload Summary

Qubits	State Dim	Total Vectors	Data Size
8	256	6,400	0.01 GB

Encoding Result

Method: `amplitude`

Output shape: `(64, 256)` complex64

Run Encode

Sample latency (8q): 0.0970 ms/vector

Est. Throughput

10,309 vec/s

(a) Pipeline Exploration

QDP: GPU-Accelerated Quantum Data Processing

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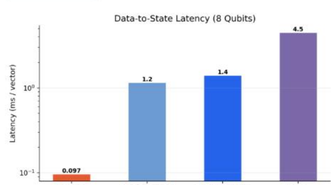
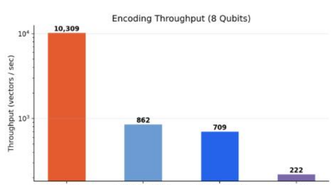
Mahout QDP vs PennyLane vs Qiskit

Run Benchmark





vs PL default	vs PL lightning	vs Qiskit
12x	15x	46x

(b) Framework Comparison

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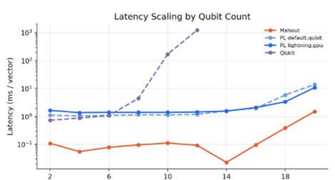
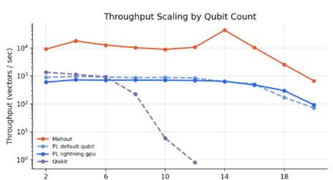
Performance Scaling by Qubit Count





Speedup Summary (sample data)

Qubits	Mahout (ms)	PL default (ms)	PL lightning (ms)	Qiskit (ms)	vs PL default	vs PL lightning
2	0.110	1.12	1.67	0.7	10x	15x
4	0.056	1.05	1.38	0.9	19x	25x

(c) Scaling Analysis